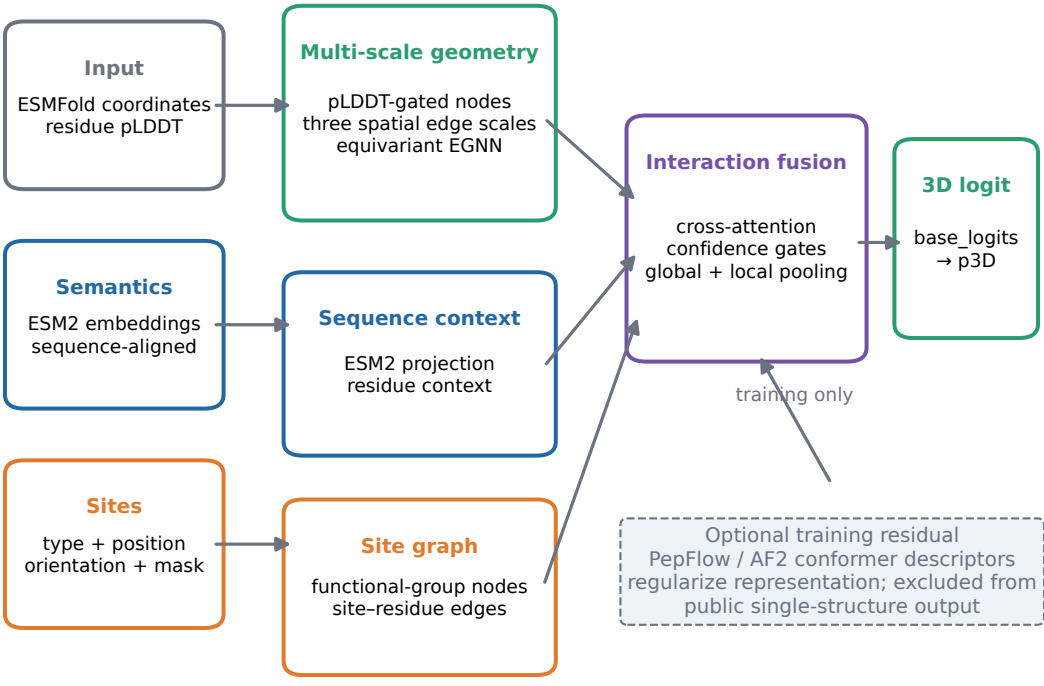
pLDDT-aware multi-scale structural expert
Input
ESMFold coordinates
residue pLDDT
Multi-scale geometry
pLDDT-gated nodes
three spatial edge scales
equivariant EGNN
Semantics
ESM2 embeddings
sequence-aligned
Sequence context
ESM2 projection
residue context
Sites
type + position
orientation + mask
Site graph
functional-group nodes
site–residue edges
Interaction fusion
cross-attention
confidence gates
global + local pooling
3D logit
base_logits
→ p3D
training only
Optional training residual
PepFlow / AF2 conformer descriptors
regularize representation; excluded from
public single-structure output